CONCEPT CHECK 54.2

1. What two components contribute to species diversity? Explain how two communities with the same number of species can differ in species diversity.
2. How is a food chain different from a food web?
3. **WHAT IF?** Consider a grassland with five trophic levels: grasses, mice, snakes, raccoons, and bobcats. If you released additional bobcats into the grassland, how would grass biomass change if bottom-up control applied? If top-down control applied?
4. **MAKE CONNECTIONS** Rising atmospheric CO_2 levels lead to ocean acidification (see Figure 3.12) and higher ocean temperatures, both of which can reduce krill abundance. Predict how a drop in krill abundance might affect other organisms in the food web shown in Figure 54.17. Which organisms are particularly at risk? Explain.

For suggested answers, see Appendix A.

CONCEPT 54.3

Disturbance influences species diversity and composition

Decades ago, most ecologists thought that biological communities are at equilibrium, a more or less stable balance, unless seriously disturbed by human activities. This "balance of nature" view focused on competition as a key factor determining the composition and stability of communities. *Stability* in this context refers to a community's tendency to reach and maintain a relatively constant composition of species.

Proponents of this view thought that the community of plants at a site had only one stable equilibrium, a *climax community* controlled solely by climate. They also argued that biotic interactions caused the species in the community to function as an integrated unit—in effect, as a superorganism. This argument was based on the observation that certain species of plants are often found together, such as the oaks, maples, birches, and beeches in deciduous forests of the northeastern United States.

Other ecologists questioned whether most communities were at equilibrium and challenged the concept of a single climax community. They thought that differences in soils, topography, and other factors created many potential communities that were stable within a region. Rather than functioning as an integrated unit, communities could be viewed as chance assemblages of species found together because they have similar abiotic requirements, such as for temperature, rainfall, and soil type. Moreover, evidence shows that disturbance keeps many communities from reaching a state of equilibrium in species diversity or composition. A **disturbance** is an event, such as a storm, fire, flood, drought, or human activity, that changes a community by removing organisms from it or altering resource availability.

This emphasis on change in communities led to the formulation of the **nonequilibrium model**, which describes most communities as constantly changing after disturbance. Even relatively stable communities can be rapidly transformed into nonequilibrium communities. Let's examine some of the ways that disturbances influence community structure.

Characterizing Disturbance

The types of disturbances and their frequency and severity vary among communities. Storms disturb almost all communities, even those in the oceans through the action of waves. Fire is a significant disturbance; in fact, chaparral and some grassland biomes require regular burning to maintain their structure and species composition. Many streams and ponds are disturbed by seasonal flooding and drying. A high level of disturbance is generally the result of frequent *and* intense disturbance, while low disturbance levels can result from either a low frequency or low intensity of disturbance.

The **intermediate disturbance hypothesis** states that moderate levels of disturbance foster greater species diversity than do high or low levels of disturbance. High levels of disturbance reduce diversity by creating environmental stresses that exceed the tolerances of many species or by disturbing the community so often that slow-growing or slow-colonizing species are excluded. At the other extreme, low levels of disturbance can reduce species diversity by allowing competitively dominant species to exclude less competitive ones. Meanwhile, intermediate levels of disturbance can foster greater species diversity by opening up habitats for occupation by less competitive species. Such intermediate disturbance levels rarely create conditions so severe that they exceed the environmental tolerances or recovery rates of potential community members.

The intermediate disturbance hypothesis is supported by many terrestrial and aquatic studies. In one study, ecologists in New Zealand compared the richness of invertebrates living in the beds of streams exposed to different frequencies and intensities of flooding **(Figure 54.24)**. When floods occurred either very frequently or rarely, invertebrate richness was low. Frequent floods made it difficult for some species to become established in the streambed, while rare floods resulted in species being displaced by superior competitors. Invertebrate richness peaked in streams that had an intermediate frequency or intensity of flooding, as predicted by the hypothesis.

Although moderate levels of disturbance appear to maximize species diversity in some cases, small and large disturbances also can have important effects on community structure. Small-scale disturbances can create patches of different habitats across a landscape, which help maintain diversity in a community. Large-scale disturbances are also a natural part of many communities. Much of Yellowstone National Park, for example, is dominated by lodgepole pine, a tree species that requires the rejuvenating influence